

Week 1 • Day 4

Exploratory Data Analysis — Part 1

Created 5 visualisation charts revealing key churn patterns: distributions, correlations, and contract insights

DATE	INTERN NAME	PROJECT	MENTOR
Day 4 of 28 Week 1		Customer Churn Prediction & LTV Engine	

■ Technologies Used Today

Python

Pandas

Seaborn

Matplotlib

NumPy

PostgreSQL

■ Today's Key Metrics

5

Charts Created

42.7%

M-t-M Churn Rate

0.83

TotalCharges-Tenure Corr.

Avg Churner Monthly Bill

■ Day Objectives

- Understand the purpose of EDA and visual analysis
- Create churn distribution chart (pie + bar)
- Build tenure distribution histogram by churn status
- Visualise monthly charges distribution and boxplot
- Create churn rate by contract type bar chart
- Generate numeric feature correlation heatmap

■ Tasks Completed Today

1. Created notebooks/eda_part1.py — complete EDA visualisation script
2. Loaded cleaned dataset from PostgreSQL and converted TotalCharges to numeric
3. Chart 1: Churn distribution — pie chart (73.5%/26.5%) + count bar chart with annotations
4. Chart 2: Tenure distribution — overlapping histograms by churn status + boxplot comparison
5. Chart 3: Monthly charges — histogram with hue=Churn + boxplot showing medians
6. Chart 4: Churn rate by contract type — bar chart with percentage labels (42.7%, 11.3%, 2.8%)
7. Chart 5: Correlation heatmap — numeric features including binary churn target
8. Saved all 5 charts as PNG files at 150 DPI to reports/figures/
9. Documented key insights from each chart in notebook comments
10. Committed all chart files and scripts to GitHub

■ Concepts Learned

- EDA: why we explore data visually before building any model
- Seaborn vs Matplotlib: Seaborn for statistical plots, Matplotlib for customisation
- Class imbalance: 26.5% churn means models need special handling
- Correlation coefficient: +1 = perfect positive, -1 = perfect negative, 0 = none
- Boxplot: shows median, Q1, Q3, and outliers — better than histograms for comparison
- Why high monthly charges correlate with churn (customers feel overcharged)
- How to save publication-quality charts with `bbox_inches='tight'`

■ Tools / Software Used

Tool / Library	Version / Notes	Status
Seaborn	0.13.0	✓ In use
Matplotlib	3.8.2	✓ In use

■ Outputs / Deliverables Produced

- ◆ reports/figures/01_churn_distribution.png — pie + bar chart
- ◆ reports/figures/02_tenure_by_churn.png — histogram + boxplot
- ◆ reports/figures/03_monthly_charges.png — distribution comparison
- ◆ reports/figures/04_churn_by_contract.png — churn rate bar chart
- ◆ reports/figures/05_correlation_heatmap.png — feature correlations
- ◆ notebooks/eda_part1.py — complete visualisation script
- ◆ GitHub commit with 5 chart images and script

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
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Charts not displaying properly — blank window appeared	Added plt.show() after plt.savefig() — save first, then display
TotalCharges had NaN after pd.to_numeric conversion	Used errors='coerce' in pd.to_numeric then filled 11 NaN with MonthlyCharges values

■ GitHub Commit Record

Commit Message	Week1-Day4: EDA Part 1 — 5 visualisation charts, key churn insights documented
Branch	main
Files Changed	notebooks/eda_part1.py, reports/figures/01-05 PNG charts
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ 5 charts saved to reports/figures/	■ Churn distribution chart created
■ Tenure histogram by churn done	■ Monthly charges chart done
■ Contract churn rate chart done	■ Correlation heatmap done
■ All charts at 150 DPI	■ Key insights documented
■ eda_part1.py committed	■ Day 4 commit on GitHub

■ Self Assessment

9/10 Self Rating	7 Hours Spent	Week 1, Day 4 Progress	4/28 Day in Program
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■ Key Learnings & Reflections

- ★ Churners have significantly shorter tenure — most leave in the first 12 months
- ★ Monthly charges are higher for churners (\$74 avg vs \$61 for retained)
- ★ TotalCharges and tenure are strongly correlated (0.83) — expected, not a bug

★ Always save charts before plt.show() to avoid blank output

■ Plan for Tomorrow

- Deep-dive into categorical features: internet service, payment method, add-ons
- Create churn rate comparison for all 6 add-on services
- Analyse senior citizen churn rate vs non-senior
- Save 4 more charts (06-09)

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED